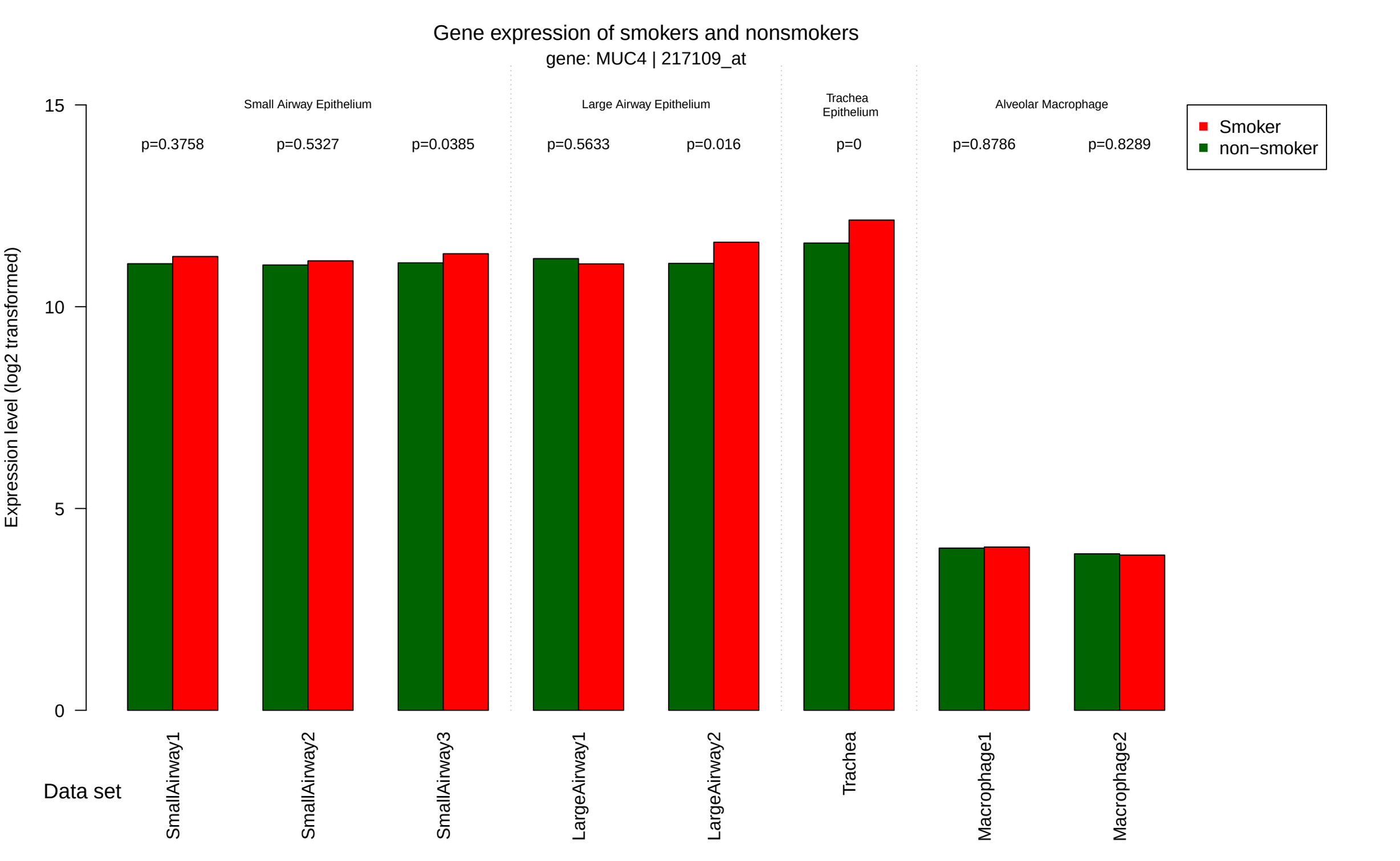
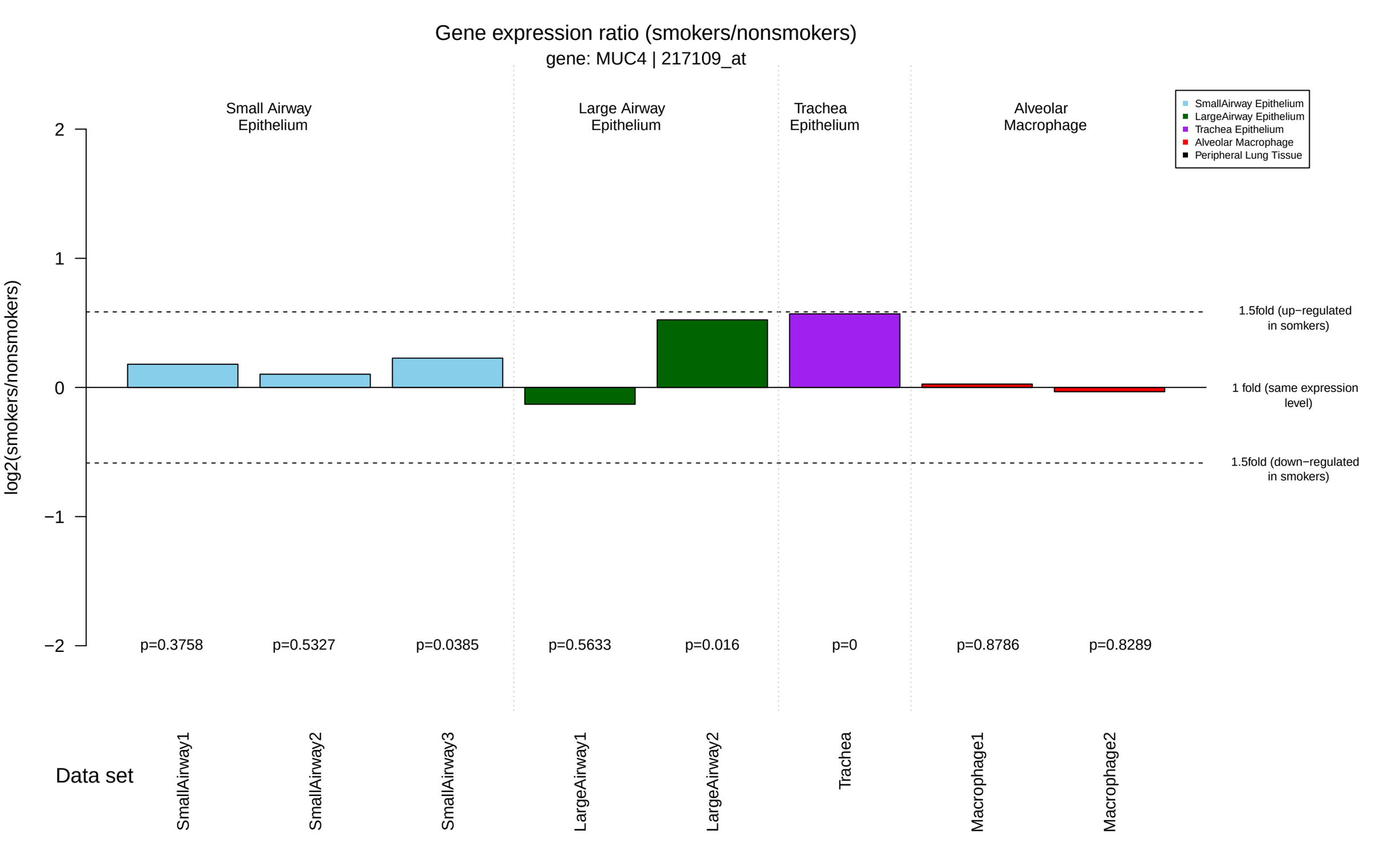
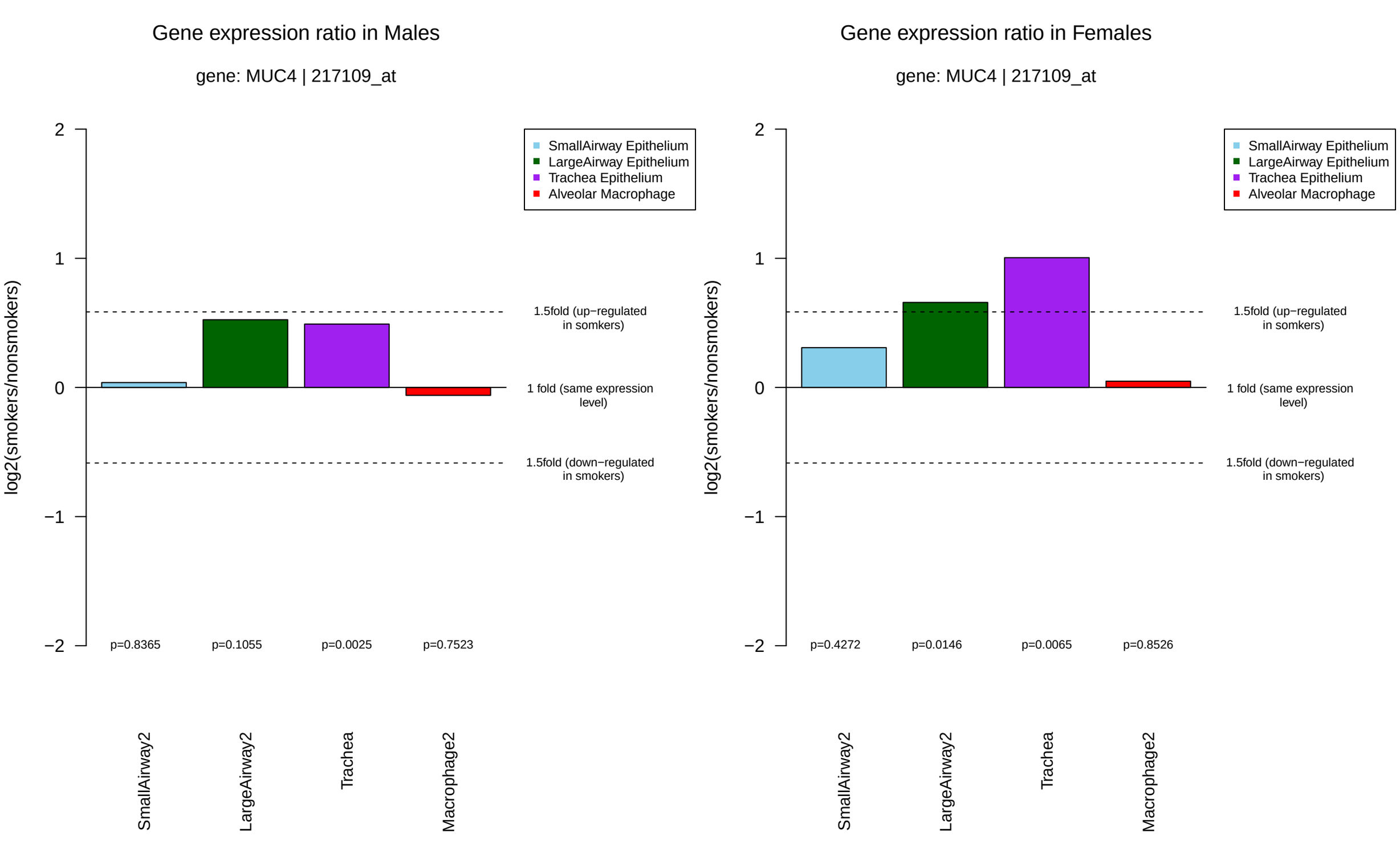
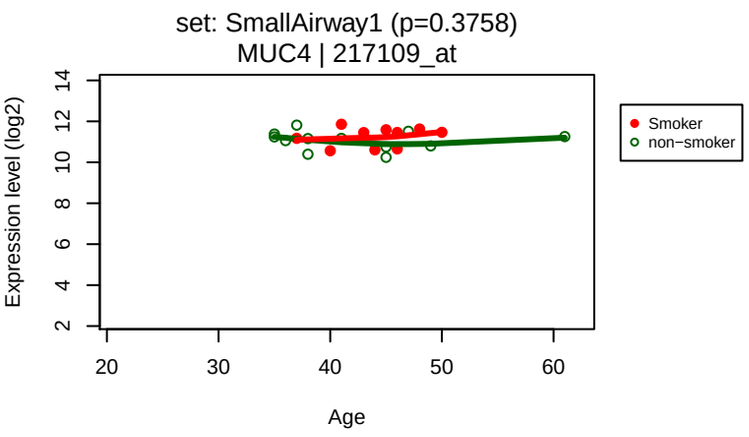
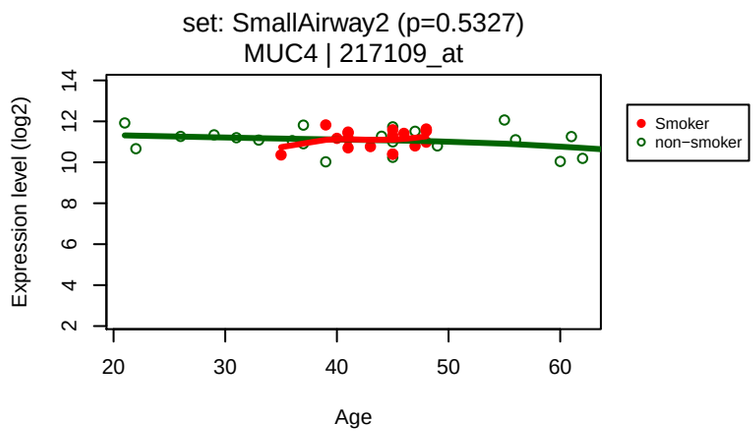
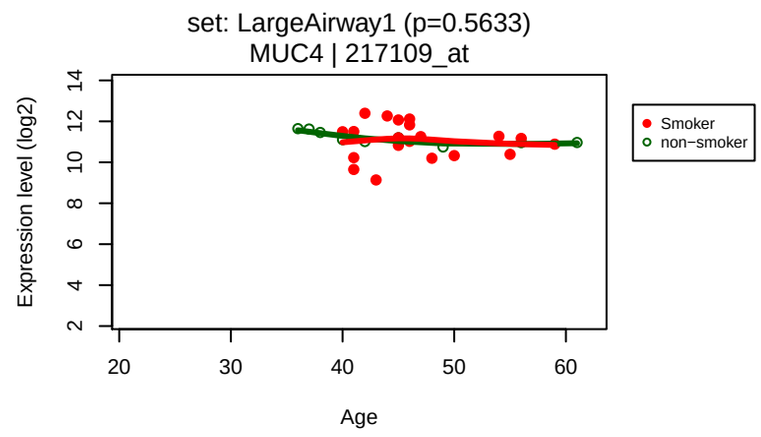

Gene expression of smokers and nonsmokers
gene: MUC4 | 217109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3758
p=0.5327
p=0.0385
p=0.5633
p=0.016
p=0
p=0.8786
p=0.8289
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

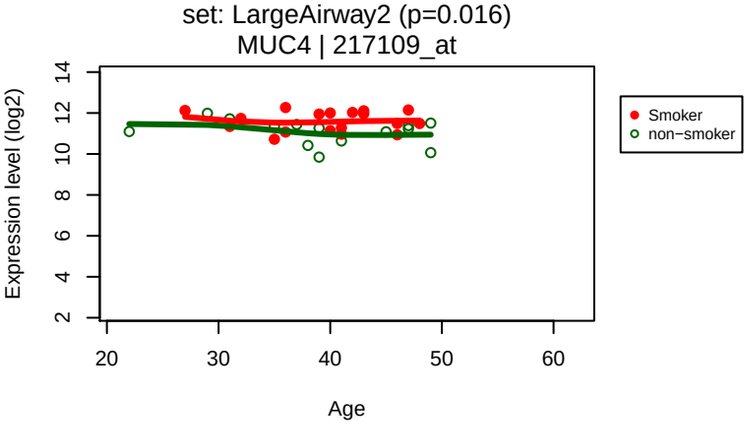

Gene expression ratio (smokers/nonsmokers)
gene: MUC4 | 217109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3758
p=0.5327
p=0.0385
p=0.5633
p=0.016
p=0
p=0.8786
p=0.8289
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

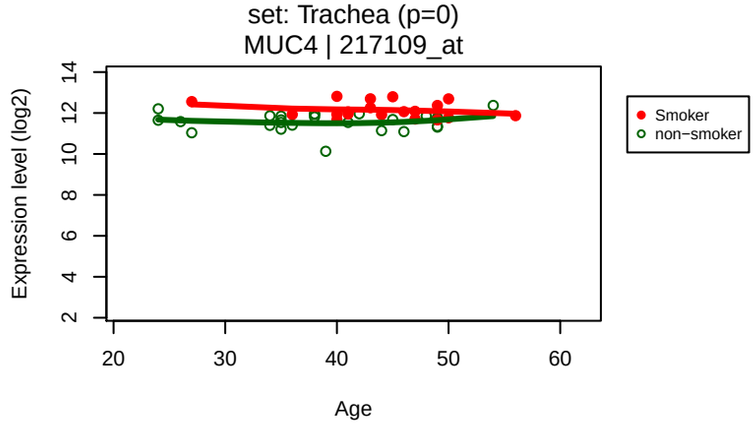

Gene expression ratio in Males
gene: MUC4 | 217109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8365
p=0.1055
p=0.0025
p=0.7523
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC4 | 217109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4272
p=0.0146
p=0.0065
p=0.8526
SmallAirway2
LargeAirway2
Trachea
Macrophage2

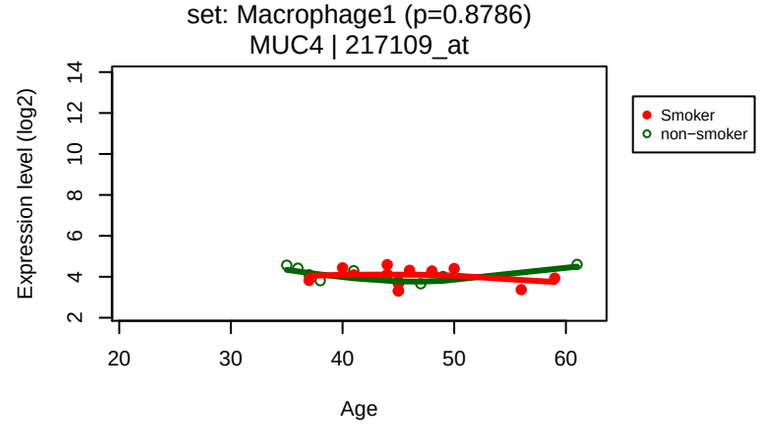

set: SmallAirway1 (p=0.3758)
MUC4 | 217109_at
Expression level (log2)
Age
Smoker
non−smoker

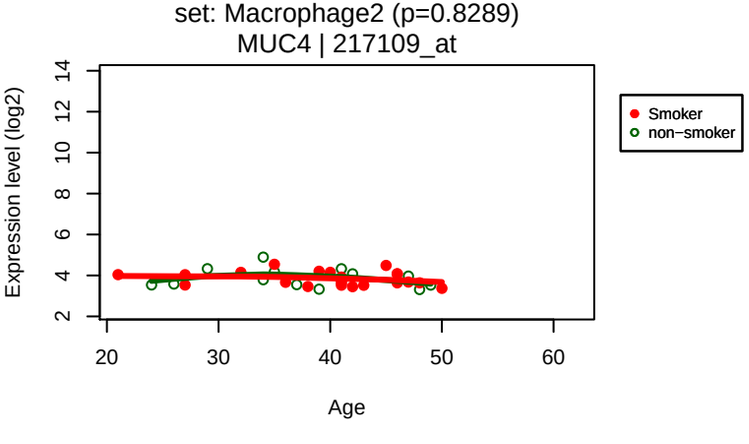

set: SmallAirway2 (p=0.5327)
MUC4 | 217109_at
Expression level (log2)
Age
Smoker
non−smoker

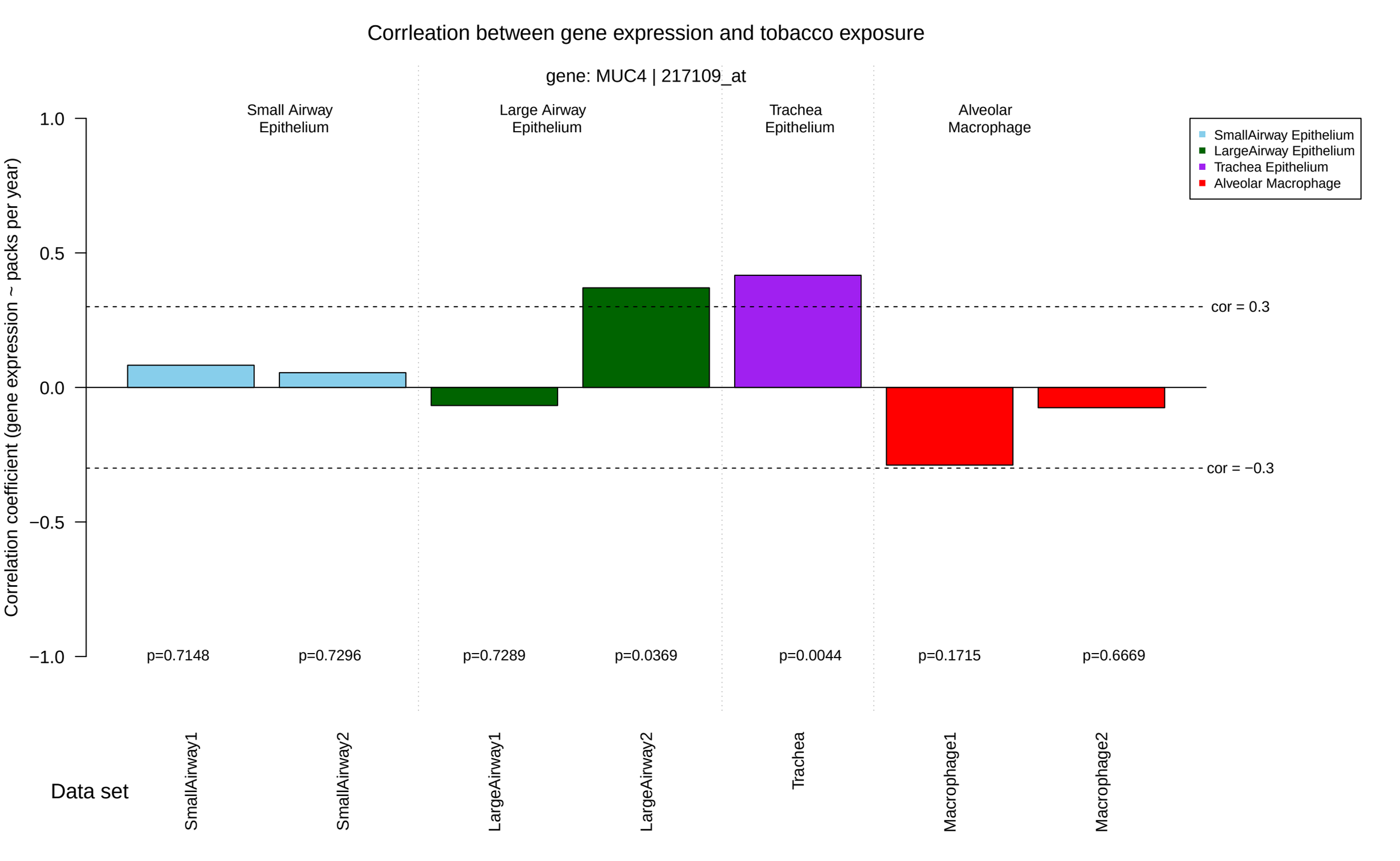

set: LargeAirway1 (p=0.5633)
MUC4 | 217109_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.016)
MUC4 | 217109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
MUC4 | 217109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8786)
MUC4 | 217109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8289)
MUC4 | 217109_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC4 | 217109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7148
p=0.7296
p=0.7289
p=0.0369
p=0.0044
p=0.1715
p=0.6669
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2